

BCD Version Difference

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Abstract

In the domain of Automated Canary Analysis, comparative anomaly detection evaluates the stability of a new software deployment against a historical production baseline. Because microservice telemetry is inherently non-stationary, static thresholds are highly ineffective. This paper formalizes the mathematical frameworks behind two architectures for comparative detection. We first define V-Diff V1, an approach utilizing Density-Based Spatial Clustering of Applications with Noise, commonly referred to as DBSCAN, combined with linear regression guards. We analyze its theoretical vulnerabilities through specific operational use cases, notably partial recovery masking, global outage false positives, and transient noise. Subsequently, we introduce V-Diff V2, a robust, deterministic statistical gating pipeline. V2 leverages the Median Absolute Deviation, adaptive percentile ranking, and concurrent cross-series correlation to resolve these vulnerabilities and achieve highly accurate, explainable anomaly detection in volatile environments.

1 Introduction

Comparative anomaly detection operates on the premise that traffic patterns, resource utilization, and error rates fluctuate due to diurnal cycles and systemic load. A comparative engine models the distribution of a stable control group, representing the legacy build over a historical window that typically spans 7 days, to establish an expected behavioral envelope for the treatment group, representing the new deployment. This paper documents the algorithmic shift from spatial machine learning models, specifically V-Diff V1, to non-parametric, robust statistical filtering, specifically V-Diff V2, detailing the practical use cases that necessitated this architectural overhaul.

2 V-Diff V1: The Density-Based ML Architecture

The original V-Diff architecture framed time-series anomaly detection as an unsupervised spatial clustering problem, assuming that anomalous deployments would topologically diverge from historical norms.

During the preprocessing and normalization phase, a historical time series $X_{\text{hist}} = \{x_{\text{hist},1}, x_{\text{hist},2}, \dots, x_{\text{hist},n}\}$ and a recent evaluation window $X_{\text{eval}} = \{x_{\text{eval},1}, \dots, x_{\text{eval},k}\}$ are evaluated. Both vectors are smoothed using a rolling median filter of length w to attenuate high-frequency noise. Because spatial distance metrics such as Euclidean distance are highly sensitive to vector magnitude, the unified dataset $X_{\text{merged}} = X_{\text{hist}} \cup X_{\text{eval}}$ is normalized to the unit interval $[0, 1]$ via Min-Max scaling:

$$\tilde{x}_i = \frac{x_i - \min(X_{\text{merged}})}{\max(X_{\text{merged}}) - \min(X_{\text{merged}})} \quad (2.1)$$

The normalized spatial points are then subjected to the DBSCAN core engine [1]. Let ϵ be the search radius and N_{min} be the minimum density threshold. A point $\tilde{x}_j \in \tilde{X}_{\text{eval}}$ is classified as an anomaly or noise if it cannot be density-reached from the core points established by $\tilde{X}_{\text{hist}}$:

$$\text{IsAnomaly}(\tilde{x}_j) \iff |\{\tilde{x}_i \in \tilde{X}_{\text{merged}} : \text{dist}(\tilde{x}_j, \tilde{x}_i) \leq \epsilon\}| < N_{\text{min}} \quad (2.2)$$

To mitigate false positives generated by the clustering sensitivity, V1 applies two secondary post-processing guards. A Least Squares linear regression is fitted across X_{eval} to act as a downward trend guard. If the estimated slope $\hat{\beta}$ is negative, the alert is suppressed:

$$\hat{\beta} = \frac{\sum_{i=1}^k (t_i - \bar{t})(x_{\text{eval},i} - \bar{x}_{\text{eval}})}{\sum_{i=1}^k (t_i - \bar{t})^2} < 0 \implies \text{Suppress} \quad (2.3)$$

Additionally, a Gini coefficient [2] is calculated on zero-padded evaluation windows to suppress sparse, transient network spikes.

3 Operational Vulnerabilities of V-Diff V1

The rigidity of the V1 post-processing filters introduced severe operational blindspots across three distinct deployment scenarios.

The first scenario is the partial recovery, which yields false negatives. Consider a deployment that triggers a massive initial error spike, such as a rapid increase from 10 to 500 errors. The service partially stabilizes over ten minutes, drifting down to 300 errors, but remains structurally broken. Because the trajectory from 500 to 300 yields a negative derivative where $\hat{\beta} < 0$, V1 assumes the system is self-healing and suppresses the anomaly, yielding a critical false negative despite an unacceptably high error floor.

The second scenario is the transient needle, which yields false positives. When a momentary network blip causes an isolated error spike that immediately resolves to zero, V1 attempts to suppress the signal using the Gini coefficient. However, zero-padding the vector artificially depresses the mean $\bar{x}$, distorting the mathematical distribution and allowing highly concentrated transient spikes to bypass the filter, yielding a false positive.

The third scenario involves systemic global outages, which also yield false positives. If a shared backend database fails, both the legacy build and the new build simultaneously spike in errors. V1 evaluates the new build against the historical distribution of the old build, lacking a real-time concurrent check. It correctly identifies the deviation of the new build but falsely attributes the root cause to the deployment, resulting in a systemic false positive.

4 V-Diff V2: Adaptive Statistical Gating

To resolve the opaqueness of DBSCAN and the topological flaws of the V1 architecture, V-Diff V2 employs a deterministic pipeline grounded in robust statistics.

To begin, V2 relies on robust location and scale estimators. This eliminates reliance on the Gaussian mean, denoted by μ , and standard deviation, denoted by σ , which possess a breakdown point of 0% and are highly susceptible to outlier distortion. Instead, the baseline location is defined by the Median, denoted by $\tilde{x}_{\text{hist}}$, and the scale is defined by the Median Absolute Deviation, denoted by D_{MAD} , both possessing a 50% breakdown point:

$$\tilde{x}_{\text{hist}} = \text{Median}(X_{\text{hist}}) \quad (4.1)$$

$$D_{\text{MAD}} = 1.4826 \cdot \text{Median}(|x_{\text{hist},i} - \tilde{x}_{\text{hist}}|) \quad (4.2)$$

To prevent infinite scaling in perfectly flat metrics, the robust scale estimator S_{hist} is floored dynamically to account for minimal variations:

$$S_{\text{hist}} = \max(D_{\text{MAD}}, 0.1 \cdot \tilde{x}_{\text{hist}}, 1.0) \quad (4.3)$$

To address volatile ceilings, V2 implements volatility profiling and an adaptive rank gate. Resource metrics often hover near zero but experience routine bursts to a hard limit. V2 dynamically tunes its sensitivity based on the historical volatility of the control group. The Coefficient of Variation, denoted by C_v , is defined robustly as follows, provided that $\tilde{x}_{\text{hist}} > 0$:

$$C_v = \frac{D_{\text{MAD}}}{\tilde{x}_{\text{hist}}} \quad (4.4)$$

The empirical cumulative distribution function of the historical baseline, denoted by $\hat{F}_{\text{hist}}$, is evaluated at the current target value $x_t \in X_{\text{eval}}$. To pass the rank gate, the target must strictly exceed a dynamic percentile threshold α , which tightens automatically for highly volatile signals:

$$\alpha = \begin{cases} 0.999 & \text{if } C_v > 0.8 \\ 0.990 & \text{otherwise} \end{cases} \quad (4.5)$$

$$\hat{F}_{\text{hist}}(x_t) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{x_{\text{hist},i} \leq x_t\}} \geq \alpha \quad (4.6)$$

To address global outages, V2 utilizes a correlated suppression check. To distinguish local deployment regressions from systemic infrastructure events, V2 executes a cross-series concurrency evaluation. If both the control and treatment groups experience a simultaneous surge surpassing an absolute severity

threshold C_{surge} at time t , and their magnitude ratio remains bounded below a correlation constraint R_{max} , the anomaly is suppressed:

$$\text{Suppress} \iff (x_{\text{hist},t} \geq C_{\text{surge}}) \wedge (x_{\text{eval},t} \geq C_{\text{surge}}) \wedge \left(\frac{x_{\text{eval},t}}{x_{\text{hist},t}} < R_{\text{max}} \right) \quad (4.7)$$

To address partial recoveries, V2 applies a robust standard score threshold. The primary severity evaluation calculates a standardized deviation metric. Given a critical anomaly threshold Z_{crit} , the gate evaluates:

$$Z = \frac{x_{\text{eval},t} - \tilde{x}_{\text{hist}}}{S_{\text{hist}}} > Z_{\text{crit}} \quad (4.8)$$

Crucially, V2 entirely abandons the negative-slope regression guard from V1. If a system partially recovers but settles at an error plateau that still violates the threshold Z_{crit} , the anomaly mathematically persists.

Finally, to address transient needles, V2 incorporates a zero-trimmed Gini filter. To correctly filter transient density bursts without the mathematical distortion of zero-padding, the evaluation window subset $X'_{\text{eval}} = \{x_i \in X_{\text{eval}} : x_i > 0\}$ is extracted. For the sorted, strictly positive subset $x'_{(1)} \leq x'_{(2)} \leq \dots \leq x'_{(m)}$, the Gini index G is computed [2]:

$$G = \frac{2 \sum_{i=1}^m i x'_{(i)}}{m \sum_{i=1}^m x'_i} - \frac{m+1}{m} \quad (4.9)$$

The signal is suppressed if the index exceeds the maximum sparsity tolerance γ_{max} , specifically when $G > \gamma_{\text{max}}$, correctly identifying the burst as non-sustained noise.

5 Conclusion

The architectural transition from V-Diff V1 to V2 represents a critical paradigm shift from opaque, topology-based machine learning to context-aware statistical gating. By defining theoretical failure models, specifically partial recoveries, systemic outages, and transient noise, and addressing them explicitly via robust estimators, adaptive percentiles, and cross-series correlation matrices, V-Diff V2 achieves superior precision and recall characteristics. Furthermore, it ensures deterministic explainability, a foundational requirement for trust and remediation in automated continuous deployment pipelines.

References

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